

More about qubit

In classical computing, the basic unit of information is bit. A bit stores either of binary values 0 and 1. In quantum computing, the phenomenon of quantum superposition is used as the basic unit of information called 'qubit'. A qubit stores either 0 or 1 but unlike classical bit, it exists in simultaneous states of 0 and 1. That means qubit may simultaneously be 0 as well as 1, but it does not mean that it carries more information than the classical bit. When measured, qubit is forced into one of the states, 0 or 1. A qubit may manifest itself as nuclear spin of an atom, photon with horizontal or vertical polarisation, or electron with spin-up or spin-down.

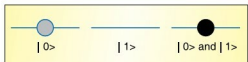


Fig. 1: Representation of qubit

different fields like biology, chemistry, computing, physics and electronics.

History of molecular electronics

The concept of molecular electronics dates back to the last century. One familiar example is the use of organic materials in displays. The use of liquid crystals display found in watches, calculators and TV sets was historically patented over fifty years ago.

Prof. Bloor points out that "molecules exhibit great variety in their structure and properties from simple diatomic species through to very large synthetic and bio-macro molecules. It is not surprising therefore that molecules can be found that process unique combination of properties which find application in fields of electronics and opto-electronics."

This idea has stimulated work on MMEs since 1950s. The reduction of size of active electronic devices compounded problems with regard to quantum effects. At this juncture,

molecular electronics—the application of molecular materials in electronics—started exploiting some of the new advanced technologies that may be beyond the scope of the silicon chip.

Prof. Bloor explains that "the continuing development of silicon micro-electronic devices of smaller size and greater complexity has brought more compact and powerful instrumentation and computing facilities into the laboratory and office. Though silicon technology holds a dominant position, the continuing reduction in dimensions of an individual device creates problems both at the fundamental and systems level. On one hand, quantum effects must ultimately come into play, dissipation and the design of testable architectures are already with us. These pressures lead inevitably to a search for alternatives to current technology that can offer prospects for realisation of devices with even higher densities of active components. MSE is one avenue which is being explored with these targets in mind."

The research and the interest in molecular electronics were mainly initiated by the late Forest Carter who conducted a series of international

Various definitions of quantum computer

A quantum computer is defined as a computational machine that makes direct use of distinctively quantum mechanical properties like superposition and entanglement to perform operation on data.

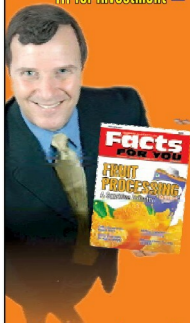
Another definition says that whereas a classical computer follows the well understood laws of classical physics, a quantum computer attempts to harness quantum mechanical phenomenon to design a fundamentally different mode of information processing machine."

The key difference between a conventional computer and a quantum computer is that the basic information processing unit in the quantum computer is qubit (quantum bit), unlike logical bit of classical computers.

Grab
your copy
today

Look Before You Leap

... for investment



A business and economic affairs magazine that influences corporate decisions on the line of
Where should we put our money next?

Launched in 1980, Facts For You updates India's top decision-makers through its regular assortment of market surveys and other important industrial information.

For subscription & advertisements, contact:



EFY Enterprises Pvt Ltd

D-87/1, Okhla Industrial Area, Phase-I,
New Delhi - 110020
Phone: 011-26810601/02/03
Email: info@efymag.com
Website: www.efymag.com